Nestify user guide

Version 1.0.0 · 16 October 2026 · Elton Sibara
eltonsibara1@gmail.com

Contents

1. Start a new house
2. Import a plan
3. Editing walls, doors and windows
4. Drawing shapes
5. Dots push/pull (shape anything)
6. Shape looks: solid, glass, frosted, glowing
7. Edit group (build one thing on its own)
8. Your own pictures as materials
9. Floors, stairs and rooms
10. Walk through the house
11. Structure parts, and lifting objects
12. Library: furniture, electronics, cars, trees, flowers
13. Stairs, railings, site, electrical and plumbing
14. Ground
15. Modern glass and solar panels
16. Real look
17. Pencil and Glass styles
18. Roof that follows the plan
19. Sun and shadows
20. X-ray (engineering view)
21. Photo-real pictures
22. Building rules check
23. Quantities and cost
24. Tape measure
25. Plans, council pack and pictures
26. Section view
27. Kitchen builder
28. Finishes catalogue
29. 360 pictures and walkthrough videos
30. Export the 3D model

31. Updates
32. Files
33. Working with big files

Start a new house

The first time Nestify opens, a short welcome shows the four steps (get a plan, shape it, check it, hand it over). Open it again from Help (?) → Welcome tour.

Press **New house** (the house-plus icon, top left):

- **Describe it** in your own words, for example "3 bedrooms, 2 bathrooms with en-suite, open plan, double garage and a study, about 180 m2" or "double storey, 4 bedrooms, laundry". The line underneath shows what Nestify understood as you type. Press **Build it** and you get a full house: walls, passage, doors, windows sized for daylight, room names, a garage with driveway and car, stairs for two storeys, and furniture, lights, sockets and plumbing in every room. Every generated plan passes the building rules check.
- **Starter plans**: Starter home, Family home, Large family home, Double storey.
- **Blank page** to draw your own with the wall, rectangle and line tools.

Everything it makes is ordinary walls and objects, so you edit it like any other plan.

Import a plan

Press the plan icon (top left), or drop a file on the window. Pictures (JPG, PNG, BMP, TIFF, WebP) and PDFs (every page; use the arrows to pick the one with the floor plan) are accepted.

1. **Sheet** – in a photo the paper is found and flattened automatically (perspective removed, standard paper shape kept), then squared up. If the guess is off, go to step 1 and drag the 4 cyan corners. A scan or PDF page is used as it is.
2. **Walls** – every wall is found: solid, hatched or hollow (drawn as two lines), straight or slanted. The sheet frame, title block, dimension lines, text, furniture, stairs and door swings are ignored, and the walls are joined at corners and T-junctions. Drag an end to fix it (ends that meet move together), drag on empty paper to add a missing wall, tap a wall to delete or split it.
3. **Size** – tap one wall you know (the longest is best), type its real length and press Set (or drag the ruler over any known length). Set the wall height; thickness is measured from the drawing.
4. **Build 3D** – gaps in walls become doors and windows drawn in solid walls become windows.

Editing walls, doors and windows

Tool	Key	What it does
Select	Space	Click a wall, the floor or the roof. Drag a selected wall's **grip** to move that corner – walls joined there follow.

Wall	L	Click corner after corner. Snaps to **Endpoint**, **Midpoint**, **On wall**, **Intersection**, the **red** / **green axes**, **Parallel** / **Perpendicular** and **From point**. Type a length + Enter for an exact wall. Ending on a wall splits it, so the joint stays editable.
Move	M	Drag a wall; walls joined to it stretch (Ctrl pulls it free). Drag a door/window along its wall. Type a distance + Enter afterwards for an exact move.
Rotate	Q	Drag to turn the selected wall (snaps to 15°).
Push/ Pull	P	Drag any face: the top changes height, a side changes thickness (walls) or width/ depth (objects), a wall end changes length. The face lights up orange; the opposite face stays put (Ctrl: both sides). Type a size + Enter, or + / – for 5 cm steps.
Split	S	Click a wall where it should become two.
Opening	D	Drag along a wall, then pick a door, window, passage or separate wall.

Click the orange arrow strip on the Layers or Properties panel (or press `[` / `]`) to fold it away for a bigger 3D view; on a small window they fold away by themselves. T toggles top view, F frames the house, Delete removes, Esc cancels, Ctrl+Z / Ctrl+Y undo / redo, Ctrl+S save, Ctrl+O open. The **Measurements** box (bottom right) accepts `3.5`, `3500mm`, `350cm`, `12ft`, `30in`.

Keyboard resizing of the selection: Ctrl+←/→ thinner/thicker (narrower/wider), Ctrl+↑/↓ lower/higher, Alt+←/→ shorter/longer (shallower/deeper), Shift for 1 cm steps. A tips card in the 3D view (lightbulb button) always lists the keys for what is selected.

Drawing shapes

- **Rectangle (R)** and **Circle (C)**: draw on the ground or on the top face of anything, then move the mouse up and click to pull the shape into a solid (or type `3,2` / a radius, Enter, then a height, Enter). Draw on top of a solid to stack another one. Solids are objects: Move, Rotate, Push/Pull, copy, delete.
- **Line (N)**: click corner after corner for any outline, **A** makes the next side an arc (click its end, move to bend it, click); close on the first corner or Enter, then pull it up into a solid.
- **Paint bucket (B)**: choose a material or a colour, click walls, roof, floors, rooms or solids.
- **Groups**: Ctrl+click to select several things, **Ctrl+G** groups them (move, turn, copy and delete together), **Ctrl+Shift+G** ungroups.
- **Scenes**: save the current view (camera, style, floor) and glide back to it.
- **Tags**: show or hide whole kinds of things (Furniture, Electrical, Plumbing, Garden, Shapes...).

Dots push/pull (shape anything)

The Dots tool (left rail, or **U**) shapes the solids you draw - like SketchUp's draw-on-a-face and Push/ Pull.

1. Pick the Dots tool. A card at the bottom of the 3D view asks how to join the dots: **Straight lines** or **Smooth curve** (Nestify remembers your choice).
2. Click a face of a shape (a box from Rectangle, a cylinder from Circle, a Line outline): that is dot

1.
3. Click more dots around the outline you want. They snap to corners and edges. Backspace takes one back.
4. Click the first dot again (or double-click, or Enter). The outline turns **green**: it is ready.
5. Hold the **right mouse button** and drag: **into** the face pushes the outline in (a recess, a window bay, a hole right through if you go far enough), **out** of the face pulls it out (a bump, a bay, a balcony box). The shape changes as you drag; let go to keep it. Looking straight at a face, drag up to pull and down to push. Or type a distance + Enter (negative pushes in).

If your dots go round (nearly) the whole face, the whole face moves, like Push/Pull. Every new face can be shaped again, so a modern house can start as one big box: push in the bays, pull out the overhangs. Ctrl+Z undoes a step. Esc cancels.

Shape looks: solid, glass, frosted, glowing

Select a shape you drew (Rectangle, Circle, Line, or one edited with Dots). Under **Look** on the right: **Solid**, **Glass** (clear, see-through), **Frosted** (milky, blurry glass that lets light through) or **Glow** (lit from inside; in the evening it lights the walls and ground around it, and in a photo render it is a real light). **Colour** changes the tint of the glass or the colour of the glow. Material still offers brick, wood, tiles, your own pictures and the rest.

Edit group (build one thing on its own)

Like opening a group in SketchUp: double-click a shape, a piece of furniture or a group, or select it and press **Edit group** on the right. The rest of the house turns into faint ghost lines and the camera moves in. Draw rectangles, circles and lines, push / pull them, or add Library items: everything you make joins the group, and only the group's pieces can be clicked and dragged (even behind walls). **Done** (top left) or **Esc** brings the whole house back, at the view you had before.

Your own pictures as materials

Any picture can be a material: floor tiles, marble, wood, brick, stone, wallpaper, a logo. **Material > Your own picture** on a room floor, a wall, the roof or a shape you drew: choose the picture and say how big one picture is in real life (0.6 m for a 600 mm tile). Nestify makes it repeat without seams; it shows in the Real style and in photo-real pictures. A house keeps up to 9 pictures, and they can be reused anywhere.

Shapes can also be made of **glass** (Material > Glass) - for your own windows, glass boxes and balustrades.

Floors, stairs and rooms

The bar at the top left of the 3D view lists the floors. **+ Floor** adds a storey on top (empty, with the outside walls copied, or a full copy). Click a floor to edit it; the floors above it are hidden so you can work. **All floors** shows the whole house, the roof always sits on the top floor, and ⋮ renames or

deletes a floor. Put **Stairs** (Library → Furniture) on a floor and the floor above gets a hole over them; when walking you climb them, or press **PageUp** / **PageDown** to change floor.

Rooms are found automatically from the walls (a doorway counts as closed). Each room shows its name and area in the view (the **Rooms** chip turns the labels off) and in the Layers list, with the total area at the top. Click inside a room to rename it and give it its own floor material and colour; names stay with the room when you move walls.

Walk through the house

Press the walking-person button (or **W**). You start outside the front door at eye height (1.6 m) and walk in like a visitor: **W**/**S** or ↑/↓ walk, **A**/**D** step sideways, ←/→ or **Q**/**E** turn, drag to look around, **Shift** walks faster, the wheel takes a step, and clicking the floor walks you there. Walls, windows and furniture stop you; doors swing open and passages let you through. A mini-map shows where you are (click it to go there) and a **Rooms** list jumps into each room (found automatically, with its area). **Esc** stops walking and puts the camera back where it was.

Structure parts, and lifting objects

The library has a **Structure** tab: round and square concrete columns, brick piers, steel H-columns, timber poles, concrete / steel / timber beams, lintels, roof trusses, purlins, strip footings, slabs, boundary walls, palisade fencing, a carport, a water tank, downpipes and solar panels. Resize any of them like other objects.

To lift anything off the floor (a beam, a shelf, a solar panel onto the roof): **Page Up** / **Page Down** (Shift for 1 cm steps), the **Up 10 cm** / **Down 10 cm** buttons, or **Shift + drag** it up and down. Footings start below the ground; the X-ray view shows them.

Library: furniture, electronics, cars, trees, flowers

The sofa button opens the Library: 37 items (sofas, beds, dining set, wardrobe, kitchen, bathroom, TV, fridge, washing machine, stove, computer, air conditioner, cars, SUV, pickup, trees, pine, palm, bushes, flowers, hedge, fence, bench, pool). Click one, move the mouse to place it and click to drop it. Every object can be dragged (Move), turned (Rotate), resized (Push/Pull or the numbers on the right), recoloured, renamed, copied (Ctrl+D) and saved to **My items** with its name, size and colour. **Import .obj model** adds your own 3D models. My items live in `~/.sketchit/library.json` and `~/.sketchit/models/`.

Stairs, railings, site, electrical and plumbing

- **Stairs** (Library > Building): straight, **L-shaped** and **U-shaped** with a landing and a sloping handrail; the floor above gets a stairwell and the walker climbs them, landing included. Also a steel balcony **railing** and a **glass balustrade**.
- **Sloping site**: the mountain button opens the Site card (slope %, the direction it falls, the level platform around the house). The ground is drawn sloping with shadows; **driveway, brick paving, stone path, gravel, instant lawn, timber deck and retaining wall** (Library > Garden)

follow it, cars and trees stand on it, and the walker walks on it. The estimate adds paving areas and the cut-and-fill to level the platform.

- **Electrical and plumbing points** (Library > Electrical / Plumbing): plug sockets, switches, TV/data points, outside lights, distribution board and stove isolator snap to the nearest wall at the usual height and face the room; ceiling lights and downlights hang from the ceiling; taps, garden taps, shower, geyser and floor drains. They show as plan symbols (layers ELECTRICAL / PLUMBING in the DXF) and the estimate counts each one plus wiring per electrical point and pipework per wet point.
- **Sun study**: the Sun card films shadows from sunrise to sunset (a frame every 20 minutes, time stamped) as an animated GIF, or MP4 when `imageio[ffmpeg]` is installed.

Ground

The Site card (the hills icon on the left) has a Ground button: grass, dusty soil, gravel and stones, concrete, paving bricks, sand, tar, or **your own picture**. A photo of a lawn, a driveway or a farm is evened out and made seamless, so it repeats without joins; set how wide the area in the photo is. The ground shows in the 3D view, on a sloping plot and in photo-real pictures.

Modern glass and solar panels

Windows: **Glass wall** (floor to ceiling), **Slim frame** and **Clerestory strip**; doors: **Glass slider** and **Glass stacker**. They have slim black aluminium frames split into tall panes, like modern houses. Change the frame colour of any wall's windows and doors (black, charcoal, bronze, silver, white) with **Material > Window and door frame colour** - for one wall or every wall. With a **flat** roof and a deep overhang (Roof settings) and **Natural stone cladding** from the finishes catalogue you get the modern villa look.

Solar panels (Library > Structure) lie on the roof: move one over the roof and it settles on the surface, tilted with the slope and facing down it. Ctrl+D copies it for the next panel.

Real look

The default **Real** style paints plaster, brick, stone or timber walls; wood, tile or concrete floors; tiled, metal or slate roofs; wooden doors and glass windows, under sky light on grass. Pick Material and Colour on the right for a wall, the floor or the roof. Night, Sketch, Neon wire and Blueprint are still there.

Pencil and Glass styles

Two more styles on the right: **Pencil** draws the house like a hand sketch - white paper, graphite edges and hatching on the shaded sides. **Glass** makes every face see-through with all its edges showing, so you can see how pieces fit together while you build (it keeps depth, unlike X-ray).

Roof that follows the plan

The roof is built from the real footprint: an L, T or U house gets one wing per arm, the ridges run into each other and meet in valleys, and courtyards stay open. Hip, gable, shed, flat, gambrel and mansard all work this way; slanted outlines get a true hip over the outline. Eaves overhang is measured from the outside face of the walls. While the Wall tool is active the roof steps aside so you can see the plan.

Sun and shadows

The sun button on the left opens the Sun card: time of day, month, place (city list), and which way north points on the plan. The house casts real shadows on the ground, on itself and from trees and cars, and the light turns warm in the morning and evening. **Shadows** and **Real sun** can be switched off.

X-ray (engineering view)

The X-ray card in Styles turns the house into a see-through engineering model: glowing blue surfaces, every edge drawn (hidden ones too) and a mesh of lines across walls, floors and furniture. Click any wall, object or anything you built and it glows blue while everything else turns into grey ghost lines, so you can study that one part inside the whole house. Click empty space to see everything in blue again.

Photo-real pictures

Press the camera-shutter icon (top right) or Export → Photo-real picture. Nestify hands the view you see to a physically based path tracer (Mitsuba 3): a real sun and sky for your place, day and hour; light that bounces around the rooms and comes in through the windows; glass that reflects; plaster, face brick, stone, timber floors, tiles and roof tiles as real materials. Choose:

- **Size**: Small, HD (1280 × 720), Full HD, Square.
- **Quality**: Quick (a draft in seconds), Good, Best (for printing).
- **Light**: as the Sun card, Morning, Golden hour, Evening with the lights on, Night.

Photos need a processor with AVX instructions (almost every Core i3/i5/i7 or Ryzen PC; not low-power Celeron, Pentium Silver or Atom chips). On those PCs the Photo button says so and offers Picture of this view.

The picture sharpens pass by pass while you watch, is cleaned of grain, toned like a photo, and kept in Pictures/Nestify as well as wherever you save it. Walk inside (W) for a photo of a room: from inside, the windows are open to the sun and the sky.

Lamps (ceiling lights, downlights, floor lamps, outside wall lights) glow, and in the evening (or with Lights on in the Sun card) they light their own rooms in the 3D view and are real lights in photos. **Curtains** (open, closed, and net curtains that let light through) are in the library under Furniture; generated houses get curtains on their windows. **TVs** (on a stand, on the wall, a computer screen) can show a picture or play an animated GIF: select the TV and press "Picture / GIF on the screen". It plays in the 3D view and shows in photos.

Building rules check

The clipboard button in the top bar checks the house against SANS 10400 rules of thumb: habitable rooms at least 6 m² and 2 m wide, ceilings 2.4 m (2.1 m for bathrooms and passages), windows about 10% of the floor for daylight and 5% opening for fresh air, doors 750 mm, stair risers up to 200 mm and treads 250 mm, outside walls 140 mm or more, stairs to every upper floor. Tap a finding to go to it. A first check, not an approval.

Quantities and cost

The calculator button in the top bar opens a bill of quantities worked out from the model: strip footings (600 x 230) and floor slabs in concrete, bricks (52 per m² per leaf, +5%), cement and sand for mortar and plaster, paint (2 coats), floor tiles / wood / screed per room, ceiling boards, roof tiles or sheets from the real roof surface, roof timber, gutters, and every door and window. Each line is priced; labour and contingency are added as percentages to get the total and a rate per m². **Prices** edits every rate (saved in `~/.sketchit/prices.json`). The currency follows the computer's country (Windows region setting): rand in South Africa, naira in Nigeria, pounds in the UK and so on, with the built-in prices converted at today's exchange rates as a starting point. Rates are fetched in the background from open.er-api.com (free, no account) when the app starts, kept in `~/.sketchit/rates.json` and refreshed every 12 hours; without internet the last rates are used (**Update rates** in Prices fetches them now); pick another currency at the top of Prices, either converting the numbers or only changing the symbol; **PDF** and **CSV** / **Excel** save the estimate with a room schedule. Rules of thumb for planning, not a quantity surveyor's bill.

Tape measure

E (or the tape button) measures between two clicked points, snapping like the Wall tool. Kept sizes show in the 3D view and print on the 2D plan; Ctrl+Z removes the last, the floor menu clears them.

Plans, council pack and pictures

Export (toolbar) offers:

- **2D plan PDF**: one A3 page per floor at a standard scale (1:50, 1:100 ...), with walls, doors and their swing, windows, stairs, furniture, room names and areas, dimension chains, scale bar, north arrow and a title block. Sizes are in the unit chosen at the bottom right.
- **2D plan PNG** of the floor you are on.
- **Elevations and sections**: the plan PDF continues with the four elevations (named by the compass from the north in the Sun card) and sections A-A and B-B drawn from the 3D model: clean lines, pale-blue glass, timber doors, grey roof, cut walls / slabs / roof in solid black, level marks (FFL of every floor, wall plate, ridge) and the overall height. The section lines are marked on every floor plan. "Elevations and sections only" saves just those pages.
- **DXF** for AutoCAD, LibreCAD, SketchUp and other CAD programs (layers WALLS, DOORS, WINDOWS, ROOMS, DIMS, FURNITURE; floors side by side, millimetres).

- **Picture of this view** at 2x or 3x the window size, without grips, labels or selection.
- The 3D model as OBJ or STL, as before (all floors and the roof).
- **Council submission pack** (PDF): site plan (plot boundary, building lines, street, roof outline, driveway, setback dimensions, coverage), every floor plan, elevations, sections, a 1:25 wall section (footing, DPC, slab, wall plate, roof build-up), and schedules (rooms and areas, doors, windows, checks, general notes). Every sheet carries the owner, erf and address from **Project details**, which the pack asks for the first time.
- **Share as a web page**: one .html file with the house in 3D inside it. Anyone can open it in a browser, on a phone too, with no app or internet: turn it with a finger or the mouse, Top / Front views, take the roof off, hide the upper floors, and see the room list.

Section view

The **Section** chip above the 3D view cuts the house open like a builder's section drawing. **Cut across** slices it at a height (1.2 m above the floor to start) so you look down into the rooms; **Front to back** and **Side to side** slice it upright, cutting away the half nearest you. Drag the slider to move the cut, and **Flip** to look from the other side. Cut walls show solid red, the way sections are drawn. Pictures you save keep the cut; **Off** (or the Section chip again) closes it.

Kitchen builder

Select the wall the kitchen goes against and press **Fit a kitchen on this wall** (on the right). Pick what you want: cupboards, a sink, a stove with an extractor hood and a fridge; without the fridge; with a tall pantry as well; or base cupboards only. Nestify uses the longest free stretch of the wall (doors and low windows are kept clear, up to 5.4 m), sizes the cupboards to fill it exactly, and leaves wall cupboards out under windows. The kitchen is one group: drag it to move it all, or Ungroup to change single units. The pieces are also in the **Kitchen** section of the Library.

Finishes catalogue

Select a room floor, a wall or the roof and press **Material**: at the top is **Choose a product**, a list of real finishes with their prices: floor tiles, porcelain, laminate, vinyl plank, solid wood, polished concrete, carpet tiles; paint colours, face brick, stone and timber cladding; concrete and clay roof tiles, IBR, corrugated and Klip-lok sheeting, slate. The 3D view takes the product's look, and **Quantities and cost** prices it on its own line (with waste). Prices follow your price list's currency. Add your own products in `~/.sketchit/materials.json`.

360 pictures and walkthrough videos

Export > 360 picture of a room: pick a room (or where you stand in walk mode) and Nestify saves a 360 photo from eye height. Post it on Facebook or open it on a phone and turn around in it.

Export > Walkthrough video makes an MP4: a slow flight around the house, a glide through your saved Scenes, then a look around every room with its name and size. Choose standard, HD or full

HD.

Export the 3D model

Export > 3D model in colour: a **.GLB** (glTF) file opens in Blender, SketchUp (glTF extension), Windows 3D Viewer and 3D web viewers; **.OBJ** comes with a **.MTL** colour file beside it. Both include the roof, doors, windows and furniture, in metres. STL (for 3D printing) is still there.

Updates

Once a day Nestify checks the developer's website for a newer version and offers the download (Help > Updates checks now). Copies from the Microsoft Store are updated by the Store instead.

Files

Save writes a `.json` project (atomically). Unsaved work is autosaved every 45 s and offered back after a crash. Export writes the 2D plan (PDF, PNG, DXF), a picture of the view, or the 3D model as OBJ (SketchUp, Blender) or STL (3D printing), in millimetres. Floors, rooms and the sun are saved in the project; older projects open as one floor.

Working with big files

The Layers list is updated in place instead of rebuilt (it used to take ~0.4 s per click), the roof is not rebuilt on every mouse move while dragging, and big changes (roof type, new plan, scaling, settings) are built in the background behind a "please wait" card.